2 CREATING ALGORITHMS

In this section you will learn how to create algorithms to solve particular problems using the constructs of sequence, selection and iteration. You will also practise displaying algorithms in flowcharts and pseudocode.

LEARNING OBJECTIVES

- Understand how to create an algorithm to solve a particular problem
- Make use of programming constructs (sequence, selection and iteration) and use appropriate conventions (flowchart, pseudocode, written description, draft program code)

ALGORITHMS FOR COMPUTERS

SUBJECT VOCABULARY

construct a smaller part from which something is built. Letters and numbers (i.e. a to z and 0 to 9) are the constructs we use to build our language and convey meaning. Bricks and cement are the basic constructs of a building

selection a construct that allows a choice to be made between different alternatives

iteration a construct that means a process is repeated. An action is repeated until a **condition** is met or a particular outcome is reached. It is often referred to as a 'loop'

GENERAL VOCABULARY

ambiguous when a statement or command does not have one obvious meaning but can be interpreted in different ways

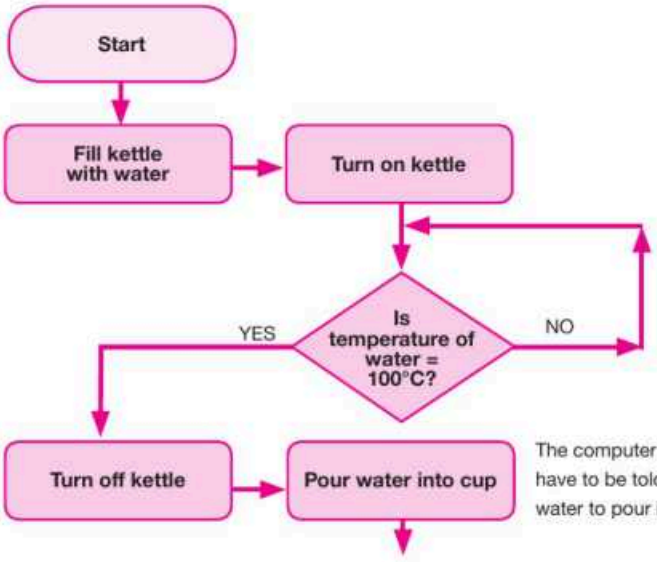
condition something that must happen before something else can happen

There was an **ambiguous** statement in the algorithm for making a cup of coffee. After filling the kettle with water and adding coffee to the cup, the next instruction was 'Wait for water to boil'.

A human can understand that this instruction means they have to keep checking the kettle over and over again until the water is boiling, but a computer is unable to understand an instruction like this in the same way. It would just wait. And wait. Forever.

Even worse, the algorithm didn't state clearly how to tell the water was boiling. Through experience, we humans assume the water is boiling when there is lots of steam, sound and bubbles; or, even better, when the kettle turns itself off. An algorithm for a computer would have to state that it must wait until the water reached 100°C.

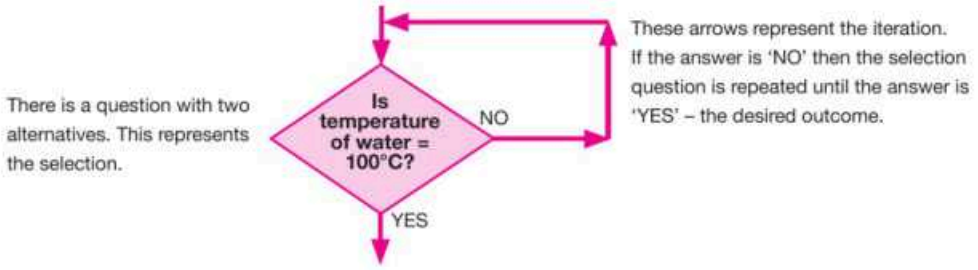
A version of this part of the algorithm, suitable for a computer, is shown in Figure 1.6. This example introduces two new **constructs** from which algorithms are created. We have already met the construct sequence – step-by-step instructions in the correct order. To add to this, we now have **selection** and **iteration**.



▶ **Figure 1.6** Part of an algorithm suitable for a computer for making coffee

REPRESENTING SELECTION AND ITERATION IN A FLOWCHART

Selection and iteration are represented in a flowchart as shown in Figure 1.7.



▶ **Figure 1.7** Selection and iteration in a flowchart

DID YOU KNOW?

We use iteration in our daily lives whenever we carry out an action over and over again. For example, at mealtimes we keep on eating until our plate is empty or we have had enough to eat.

When we're travelling by car and the traffic lights are red, we have to keep waiting until they change to green.

An actor repeats their lines over and over again, learning more each time until they know them all.

SKILLS REASONING, PROBLEM SOLVING

ACTIVITY 7

GUESSING GAMES

A student is creating a guessing game. A player has to enter a number no greater than 10. If it is too high, they are informed that they have made an error. But if it is within the range 1 to 10, they are told whether or not they have guessed the correct number. (Assume that the correct number is 3.)

Can you make an algorithm to solve this problem and express it as a written description and a flowchart?

Compare your solution to others in your group.

Check that they are correct and would produce the correct outcome.

Are some of the algorithms more efficient than others? Do they use fewer commands?

SKILLS PROBLEM SOLVING, ANALYSIS

ACTIVITY 8

CALCULATING GRADES

A school uses this algorithm to calculate the grade that students achieve in end-of-topic tests.

```
RECEIVE testScore FROM KEYBOARD
IF testScore >= 80 THEN
    SEND 'A' TO DISPLAY
ELSE
    IF testScore >= 70 THEN
        SEND 'B' TO DISPLAY
    ELSE
        IF testScore >= 60 THEN
            SEND 'C' TO DISPLAY
        ELSE
            IF testScore > 0 THEN
                SEND 'D' TO DISPLAY
            ELSE
                SEND 'FAIL' TO DISPLAY
```

```
            END IF
        END IF
    END IF
END IF
```

What would be the output of this algorithm for these test scores: 91, 56 and 78?

ITERATION

When writing programs, it is often necessary to repeat the same set of statements several times. Instead of making multiple copies of the statements, you can use iteration to repeat them. The algorithm for making a cup of coffee includes an instruction to keep waiting until the water in the kettle boils.

CHECKPOINT

Strengthen

SKILLS CRITICAL THINKING, DECISION MAKING

S1 How are sequence, selection and iteration used in algorithms? Give examples to justify your answer.

Challenge

SKILLS PROBLEM SOLVING

C1 Develop an algorithm using a flowchart that asks the user to enter their height (in metres) and weight (in kilograms) and displays their body mass index (BMI). The formula for calculating BMI is weight/height^2.

SKILLS PROBLEM SOLVING

C2 Develop an algorithm expressed as a flowchart to control the heating in a house. A thermostat monitors the temperature within the house. During the week the temperature should be 20°C between 06.00 and 08.30 in the morning and between 17.30 and 22.00 at night. At weekends it should be 22°C between 08.00 and 23.00. If the temperature in the house falls below 10°C at any time the boiler is switched on.

How confident do you feel about your answers to these questions? If you're not sure you answered them well, try the following activities again.

- For S1 have a look at the Subject vocabulary sections in 'Understanding algorithms' and 'Creating algorithms'.

SUMMARY

- The constructs sequence, selection and iteration are the basic building blocks of algorithms.